Compositions of Rigid Motions and Symmetry

Answer Key

Geometry

Quick Answers

1. 5, 1
2. 5, 5
3. 3, 2.5, -2
4. 6
5. 0, 0
6. 6, 6
7. -7, -2
8. 6, 60
9. 3, 1.5, 3, 1.5
10. 2, 3

Curriculum

Level: Geometry
HSG-CO.A – *problems 1, 4, 5, 7, 8, 9, 10*
HSG-CO.B – *problems 2, 3, 6*
Difficulty 1–5 of 5 across 22 tagged checks.

Common wrong answers

4. *If they got 3: gave the distance between the two mirror lines instead of the distance the point travelled, which is twice as far*
8. *If they got 180: halved the full turn because the hexagon looks the same after a half turn, missing the smaller angle that also works*

What this sheet is teaching, and what to listen for. The sheet builds one argument in three stages: *do* a composition (1–3), *name* the single transformation it amounts to (4–6), and then use that machinery to describe a figure's own symmetries (8–10). The commonest error is not arithmetic, it is order — "reflect then translate" and "translate then reflect" generally land in different places, and problem 7 asks the student about exactly that. When a description is wrong, check whether it names all of kind, size and centre or direction; a rotation without a centre is not an answer.

1. One point, two steps. $A(3,4)$, reflect in the x-axis then translate by $\langle 2,5\rangle$.

Reflecting in the x-axis keeps the x-coordinate and negates the y-coordinate: $A' = (3,-4)$. Translating adds the vector componentwise: x goes to $3+2 = \boxed{5}$ and y goes to $-4+5 = \boxed{1}$, so $A'' = (5,1)$.
Do one step at a time, and plot both. A student who jumps straight to the final point usually applies the translation to the original A and lands on $(5,9)$. Plotting A' makes that impossible to do by accident.

2. A segment through a composition. $P(1,1)$, $Q(4,5)$; translate by $\langle 3,-2\rangle$, then reflect in the y-axis.

Translation: $P' = (4,-1)$ and $Q' = (7,3)$.
Reflection in the y-axis negates the x-coordinate: $P'' = (-4,-1)$ and $Q'' = (-7,3)$.
Original length: $PQ = \sqrt{(4-1)^2 + (5-1)^2} = \sqrt{9+16} = \boxed{5}$.
Image length: $P''Q'' = \sqrt{(-7-(-4))^2 + (3-(-1))^2} = \sqrt{9+16} = \boxed{5}$.
The equal lengths are the point, not a coincidence. Both steps are rigid motions, so the composition is one too, and a rigid motion cannot change a distance. If a student's two lengths differ, the error is in the image coordinates — the distance formula has just caught it for them, which is a habit worth building.

3. A triangle through a composition. $A(-4,1)$, $B(-1,1)$, $C(-1,3)$; reflect in the y-axis, then translate by $\langle 0,-4\rangle$.

Reflection in the y-axis: $A' = (4,1)$, $B' = (1,1)$, $C' = (1,3)$.
Translation down 4: $A'' = (4,-3)$, $B'' = (1,-3)$, $C'' = (1,-1)$.
(a) $\overline{A''B''}$ is horizontal, so its length is just the gap in x: $4-1 = \boxed{3}$. That matches AB on the original, as it must.
(b) Midpoint of $\overline{A''C''}$: average the coordinates, $\left(\frac{4+1}{2}, \frac{-3+(-1)}{2}\right) = \boxed{(2.5,\ -2)}$.
A check that costs nothing. The midpoint of $\overline{AC}$ on the original is $(-2.5, 2)$. Reflecting that in the y-axis and dropping it 4 gives $(2.5,-2)$ — the same point. Midpoints go where the transformation sends them, so a student can verify part (b) without recomputing anything.

4. Two parallel mirrors. Reflect $P(0,2)$ in $x=1$, then in $x=4$.

(a) P is 1 unit left of $x=1$, so its image is 1 unit right of it: $P' = (2,2)$. Then P' is 2 units left of $x=4$, so $P'' = (6,2)$.
(b) $P(0,2)$ to $P''(6,2)$ is a horizontal move of $\boxed{6}$ units.
(c) (Open description — listen for all three parts of it.)
A model answer. "A translation of 6 units to the right — the vector $\langle 6,0\rangle$. Nothing is flipped: two reflections undo each other's mirror-image effect, so the final figure faces the same way as the original."
The general result behind it. The mirrors are 3 units apart and the point moved 6 units — exactly twice the gap, in the direction from the first mirror to the second. That is true for every point and every pair of parallel mirrors, and it does not depend on where the point started. Worth asking: what if the reflections were done in the other order? (A translation of 6 units to the *left*.)
Marking guide. Full credit needs the kind (translation), the size (6) and the direction (right / positive x). "A translation" alone is half an answer.
If they answered 3 in part (b): they gave the distance between the mirror lines instead of the distance the point travelled. Have them measure P to P'' on the grid.

5. Two perpendicular mirrors. Reflect $P(3,5)$ in the x-axis, then in the y-axis.

(a) Reflecting in the x-axis gives $(3,-5)$; reflecting that in the y-axis gives $P'' = (-3,-5)$.

(b) Midpoint of $\overline{PP''}$: $\left(\frac{3+(-3)}{2}, \frac{5+(-5)}{2}\right) = \boxed{(0,\,0)}$ — the origin.

(c) (Open description — listen for kind, angle and centre.)
A model answer. "A rotation of 180° about the origin — a half turn. Part (b) shows why: the origin is exactly halfway between P and its final image, and a half turn about a point is precisely the motion that sends every point to the other side of that point, the same distance away."
Marking guide. Full credit for naming the transformation, the angle (or "half turn"), and the centre. "A rotation about the origin" without the angle is incomplete — 90° about the origin sends P somewhere else entirely. "Point reflection in the origin" is also full credit; it is the same motion under another name.
Why the direction does not need stating here. A half turn clockwise and a half turn counterclockwise land in the same place, so 180° is the one rotation angle that needs no direction. Problem 6 does need one.

6. Mirrors that meet at 45°. $\triangle RST$ with $R(1,1)$, $S(5,1)$, $T(5,4)$; reflect in $y = x$, then in the x-axis.

Reflecting in $y = x$ swaps the coordinates: $R'(1,1)$, $S'(1,5)$, $T'(4,5)$.
Reflecting that in the x-axis negates the y-coordinates: $R''(1,-1)$, $S''(1,-5)$, $T''(4,-5)$.
(a) $\triangle RST$ has a horizontal leg of 4 and a vertical leg of 3, so its area is $\frac{1}{2}(4)(3) = \boxed{6}$ square units.
(b) The image has a vertical leg of 4 and a horizontal leg of 3, so its area is $\frac{1}{2}(4)(3) = \boxed{6}$ square units — unchanged, as it must be.

(c) (Open description — listen for kind, angle, direction and centre.)
A model answer. "A rotation of 90° clockwise about the origin. In coordinates the two steps send (a,b) to (b,a) and then to $(b,-a)$, and $(a,b) \mapsto (b,-a)$ is the rule for a quarter turn clockwise about the origin. Checking on T: $(5,4)$ becomes $(4,-5)$, which is what the drawing shows."
Marking guide. Full credit requires the angle *and* the direction. 90° counterclockwise is the composition done in the other order, so it is a genuinely wrong answer rather than a slip. "270° counterclockwise" is the same motion and earns full credit.
The angle between the mirrors is the tell. The lines $y = x$ and the x-axis meet at 45°, and the resulting rotation is 90° — twice the angle between them, about their intersection point. Problem 5 is the same rule with mirrors at 90° giving a 180° rotation, and problem 4 is the limiting case with mirrors that never meet.

7. Translation, then a half turn. $M(2,3)$, translate by $\langle 5,-1\rangle$, then rotate $180°$ about the origin.

Translation: $M' = (2+5,\ 3-1) = (7,2)$.
A half turn about the origin negates both coordinates: x goes to $-(2+5) = \boxed{-7}$ and y goes to $-(3+(-1)) = \boxed{-2}$, so $M'' = (-7,-2)$.
Order matters, and here is the proof. Doing the half turn first sends $M(2,3)$ to $(-2,-3)$, and translating that by $\langle 5,-1\rangle$ gives $(3,-4)$ — a different point from $(-7,-2)$. So the answer to the small question at the bottom of the problem is *no*. Composition is not commutative, and this is the cheapest example to show it.
A rotation of $180°$ about the origin is worth memorising as a rule: $(x,y) \mapsto (-x,-y)$. Students who rotate by tracing on the grid get there too, but the rule is what makes problem 5's description available.

8. Symmetry of a regular hexagon.

(a) Two families of lines: three run from a vertex to the opposite vertex, and three run from an edge midpoint to the opposite edge midpoint. That is $2 \times 3 = \boxed{6}$ lines of symmetry.
(b) Six vertices sit at equal spacings around the centre, so turning by one sixth of a full turn lands each vertex on the next one: $360 \div 6 = \boxed{\text{60 degrees}}$.
The general rule. A regular n-gon has exactly n lines of symmetry and rotational symmetry of order n, with smallest angle $360°/n$. Both parts of this problem are that one rule with $n = 6$, which is worth saying out loud so the student is not memorising the hexagon.
Why the two families look different but count the same. With an even number of sides, a vertex-to-vertex line has a vertex at each end and an edge-to-edge line has an edge midpoint at each end. With an odd number of sides every line of symmetry runs vertex to edge midpoint, so there is only one family — and still exactly n of them.
If they answered 180 in part (b): they found *a* rotation that works rather than the *smallest* one. A half turn does map the hexagon onto itself; so do $120°$, $240°$ and $300°$. The question asks for the smallest, and every other one is a multiple of it.

9. Symmetry of a parallelogram. $E(0,0)$, $F(4,0)$, $G(6,3)$, $H(2,3)$.

(a) Midpoint of $\overline{EG}$: $\left(\frac{0+6}{2}, \frac{0+3}{2}\right) = \boxed{(3,\ 1.5)}$.

(b) Midpoint of $\overline{FH}$: $\left(\frac{4+2}{2}, \frac{0+3}{2}\right) = \boxed{(3,\ 1.5)}$ — the same point.

(c) (Open explanation — listen to the reason.)
A model answer. "Both diagonals have midpoint $(3, 1.5)$, so call that point Z. A half turn about Z sends E to G and F to H, because Z is halfway along each of those segments — and it sends G to E and H to F for the same reason. Every vertex lands on a vertex, so the whole parallelogram maps onto itself: it has rotational symmetry of order 2. But it has no line of symmetry. The two slanted sides $\overline{FG}$ and $\overline{EH}$ both lean to the right rather than mirroring each other, so folding along a vertical line through Z does not bring them together, and folding along either diagonal sends E off the figure entirely."
Marking guide. Full credit needs the shared midpoint used as the centre of the half turn, plus a *reason* for the missing line symmetry — either the leaning sides or a named vertex whose mirror image is not a vertex of $EFGH$. Partial credit for the rotation argument alone. Not sufficient: "parallelograms don't have line symmetry", which is a fact recalled rather than a reason, and is in any case false for the rectangle and the rhombus.
A good follow-up. Ask which parallelograms *do* have line symmetry. A rectangle has two, a rhombus has two, and a square has four — and all of them still have the half turn, because the diagonals of every parallelogram bisect each other.

10. Why perpendicular mirrors make a half turn. Lines $x = 2$ and $y = 3$, meeting at $C(2, 3)$; $K = (4, 1)$.

(a) K is 2 units right of $x = 2$, so reflecting puts it 2 units left: $K' = (0, 1)$. Then K' is 2 units below $y = 3$, so reflecting puts it 2 units above: $K'' = (0, 5)$.

(b) Midpoint of $\overline{KK''}$: $\left(\frac{4+0}{2}, \frac{1+5}{2}\right) = \boxed{(2,\ 3)}$ — exactly the crossing point C.

(c) (Open justification — the general argument is what is being marked.)
A model answer. "Put the crossing point at the origin, so the two mirrors are the y-axis and the x-axis and a general point is (x, y). Reflecting in the vertical mirror reverses the horizontal displacement and leaves the vertical one alone: $(x, y) \to (-x, y)$. Reflecting that in the horizontal mirror does the opposite: $(-x, y) \to (-x, -y)$. So after both reflections *both* displacements from the crossing point have been reversed, and $(x, y) \mapsto (-x, -y)$ is exactly a 180° rotation about that point. Nothing in the argument used particular numbers, so it holds for every point, and therefore for every figure."
Marking guide. The question asks for an argument that covers *every* point, so re-checking K is not sufficient on its own — that is part (a). Full credit for the coordinate argument above, or for a clear description in words of the two displacements reversing one after the other, or for the midpoint observation *provided* the student says why the crossing point is the midpoint for every point and not just this one. Partial credit for a correct statement of the result with a single worked instance as the only support.
If the student is stuck: ask them to redo part (a) with a second point of their own choosing, then a third, and to say what stayed the same each time. The pattern — both displacements from C flip sign — is the argument, and most students can state it once they have seen it happen twice.